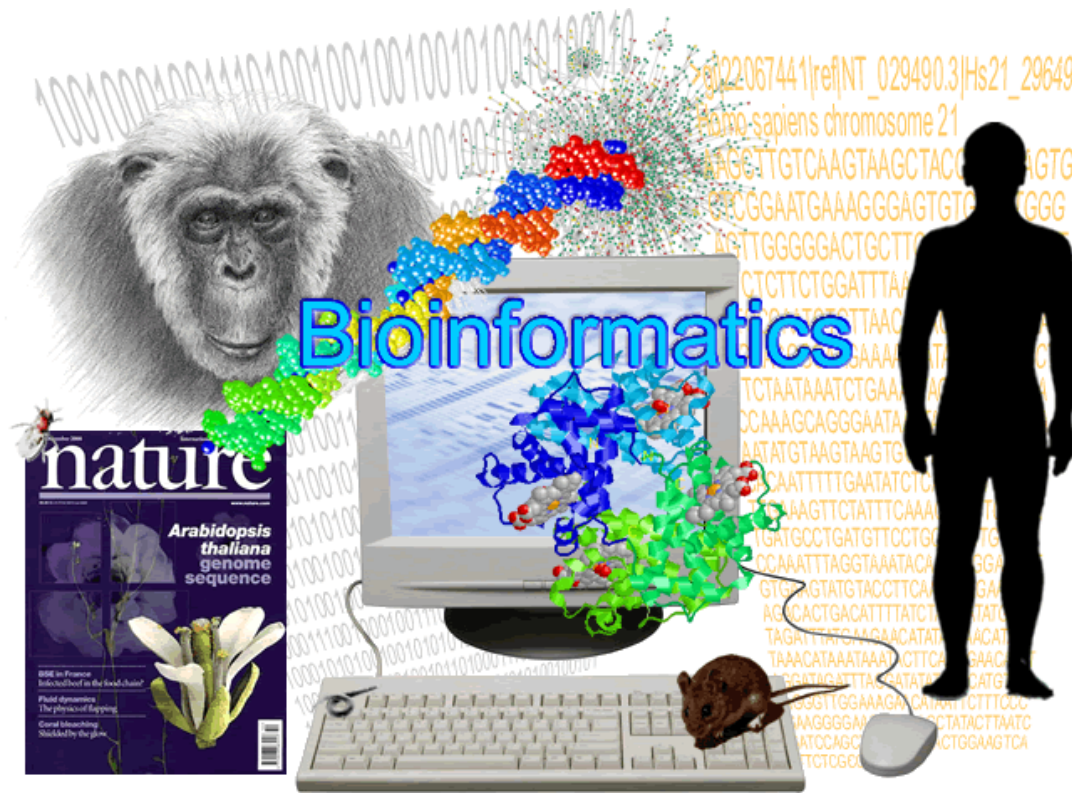


Lecture

DATABASE SEARCHING



Outline

BLAST & FASTA

Searching distantly related proteins

Scoring matrix & Statistical significance
of sequence alignment

BLAST

- BLAST (**B**asic **L**ocal **A**lignment **S**earch **T**ool) allows rapid sequence comparison of a query sequence against a database
- The BLAST algorithm is fast, accurate, and accessible both via the web and the command line

The Basic Local Alignment Search Tool (BLAST) finds regions of local similarity between sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance of matches. BLAST can be used to infer functional and evolutionary relationships between sequences as well as help identify members of gene families.

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>

BLAST: Basic Local Alignment Search Tool

Getting Started SBT WORKING SCHED...

NIH U.S. National Library of Medicine NCBI National Center for Biotechnology Information Sign in to NCBI

BLAST Home Recent Results Saved Strategies Help

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

A New version of Magic-BLAST(1.5.0) is here.

The BLAST tool for mapping large next-generation RNA or DNA sequencing runs against a whole genome or transcriptome.

Wed, 28 Aug 2019 17:00:00 EST [More BLAST news...](#)

Web BLAST



Nucleotide BLAST
nucleotide ► nucleotide



blastx
translated nucleotide ► protein



tblastn
protein ► translated nucleotide



Protein BLAST
protein ► protein

BLAST Genomes

Enter organism common name, scientific name, or tax id **Search**

Human Mouse Rat Microbes

Why use BLAST?

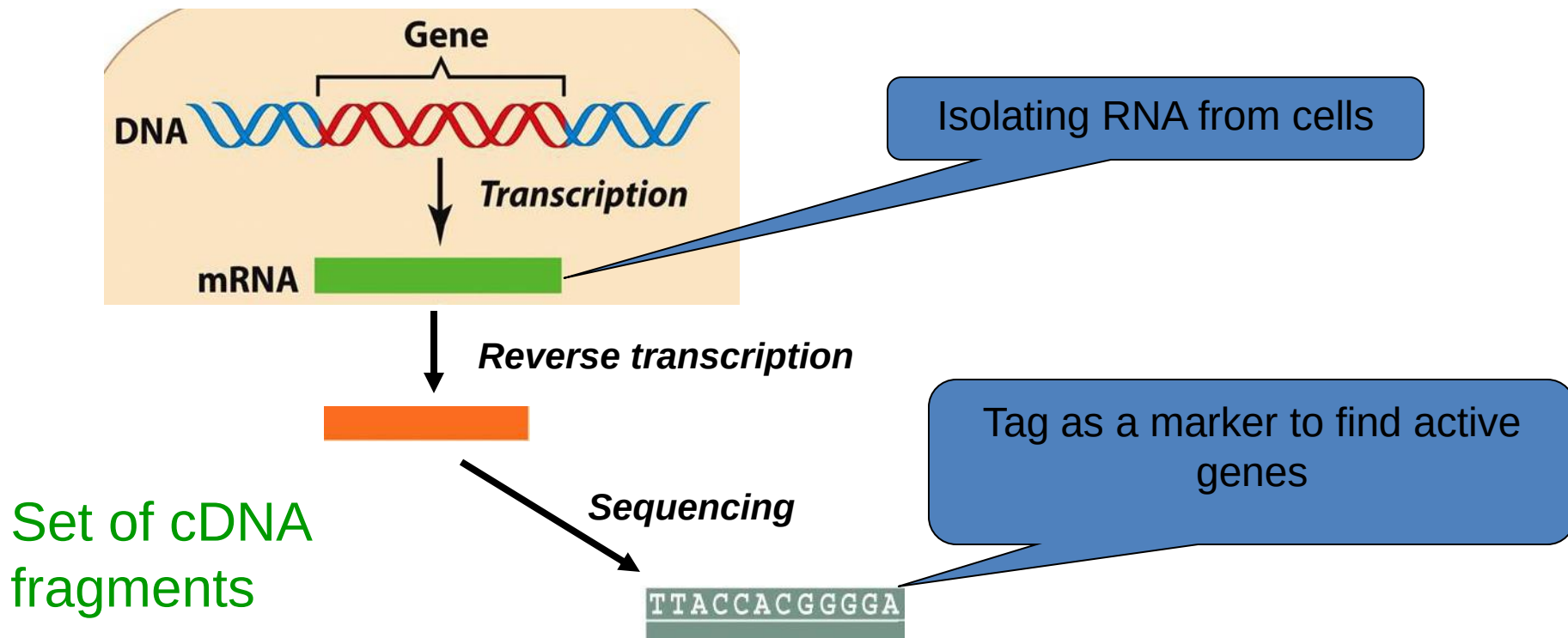
BLAST searching is fundamental to understanding the relatedness of any favorite query sequence to other known proteins or DNA sequences.

Applications include

- identifying orthologs and paralog
- discovering new genes or proteins
- discovering variants of genes or proteins
- investigating expressed sequence tags (ESTs)
- exploring protein structure and function

Expressed-Sequence Tags

- ESTs are markers associated with DNA sequences that are expressed as RNA



How a BLAST search works

- BLAST using **heuristic algorithm** allows a DNA or protein query sequence to be compared with sequences in the database
- **Main idea:**
 - Homologous sequences are likely to contain a short, high-scoring similarity region called a **word** (W) or **hit**
 - Each word gives a **seed** that triggers the alignment
 - BLAST tries to extend on both sides of the seed

How a BLAST search works

- The **Needleman–Wunsch** (1970) global alignment algorithm is not used for database searches because we are usually more interested in identifying locally matching regions such as protein domains
- The **Smith–Waterman** (1981) local alignment algorithm finds optimal pairwise alignments, but we cannot use it for database searches generally because it is too computationally intensive.

Dynamic programming algorithms

Example: for a human RBP query
...FS**GTW**YA... (query word is in green)

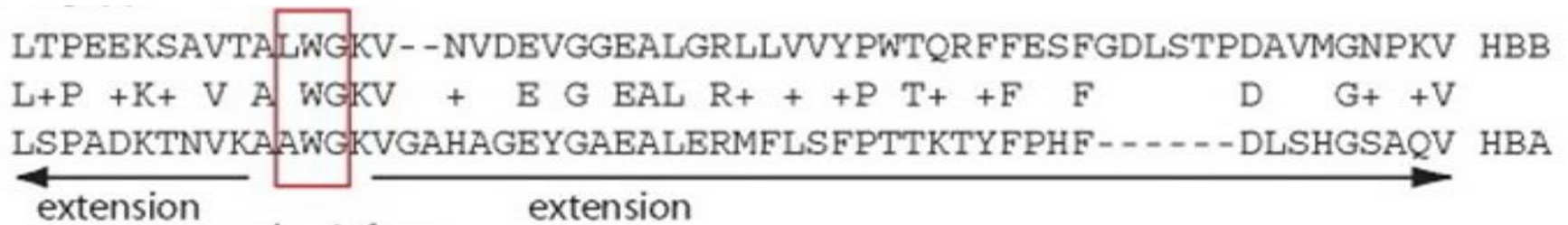
A list of words ($w=3$) is:

FSG SGT GTW TWY WYA

YSG TGT ATW SWY WFA

FTG SVT GSW TWF WYS

...



How a BLAST search works

Word size (the length of the seed) of nucleotides (blastn):

- Default: 11
- Smallest word size: 7. This increases the sensitivity of search → increase the number of positive hits

How a BLAST search works

Other options of the nucleotide BLAST:

- **Megablast** (highly similar sequences)

- Default: **28**

- Smallest word size: **16**

→ This is the best tool to find the identical match to the query sequence

- **Somewhat similar sequences** (blastn)

- Default: **11**

- **Discontinuous megablast** is optimized for more dissimilar sequences

Nucleotide BLAST: Search nucle...

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastn&PAGE_TYPE=BlastSearch&LINK_LOC=blastn

Search

NIH

U.S. National Library of Medicine

NCBI National Center for Biotechnology Information

BLAST

>> blastn suite

HomeRecent ResultsSaved Strategies

blastnblastpblasttblastntblastx

Standard Nucleotide BLAST

Enter Query Sequence

BLASTN programs search nucleotide databases using a nucleotide query. more...

Reset pageBookmark

Enter accession number(s), gi(s), or FASTA sequence(s)

Clear

Query subrange

From

To

Or, upload file

Browse...No file selected.

Job Title

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database

Human genomic + transcriptMouse genomic + transcriptOthers (nr etc.):

Nucleotide collection (nr/nt)

Organism

Optional

Enter organism name or id—completions will be suggested

exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown

Exclude

Optional

Models (XM/XP)Uncultured/environmental sample sequences

Limit to

Optional

Sequences from type material

Entrez Query

Optional

Enter an Entrez query to limit search

YouTube

Create custom database

Program Selection

Optimize for

Highly similar sequences (megablast)

More dissimilar sequences (discontiguous megablast)

Somewhat similar sequences (blastn)

Choose a BLAST algorithm

BLAST results will be displayed in a new format by default

You can always switch back to the Traditional Results page.

Windows taskbar with icons for Internet Explorer, VLC, Word, Excel, File Explorer, Firefox, PowerPoint, Chrome, and OneDrive.

How a BLAST search works

Word size (the length of the seed) of **proteins** (**blastp**):

- Default: **5**
- Smallest word size: **2**.

These parameters can be adjusted by clicking
“**Algorithm parameters**”

BLAST search steps

- Step 1: Specifying sequence of interest
- Step 2: Selecting BLAST program
- Step 3: Selecting a database
- Step 4: Selecting search parameters and formatting parameters

Step 1: input the sequence of interest

- Step 1: Specifying sequence of interest
 - Sequence can be input in **FASTA format** or as **accession number**
 - The sequence that is subject to comparison is termed the **query**
 - All sequences from the database showing similarity with the query called **subject** (or **target**)

Step 2: choose program

Program	Query	Number of database searches	Database
---------	-------	-----------------------------	----------

Protein BLAST

BLASTP	protein	1	protein
---------------	---------	---	---------

Use BLASTP to compare a protein query to a database of proteins.

Nucleotide BLAST

BLASTN	DNA	1	DNA
---------------	-----	---	-----

Use BLASTN to compare both strands of a DNA query against a DNA database.

BLASTX	DNA	6	protein
---------------	-----	---	---------

BLASTX translates a DNA sequence into six protein sequences using all six possible reading frames, and then compares each of these proteins to a protein database.

TBLASTN	protein	6	DNA
----------------	---------	---	-----

TBLASTN is used to translate every DNA sequence in a database into six potential proteins, and then to compare your protein query against each of those translated proteins.

TBLASTX	DNA	36	DNA
----------------	-----	----	-----

TBLASTX is the most computationally intensive BLAST algorithm. It translates DNA from both a query and a database into six potential proteins, then performs 36 protein-protein database searches.

Six reading frames of translation

GGT →

FRAME +3

TGG →

FRAME +2

ATG →

FRAME +1

5' – ATGGTATTTGTACAGC – 3'

Plus Strand

3' – TACCATAAACATGTCG – 5'

Minus Strand

← TCG

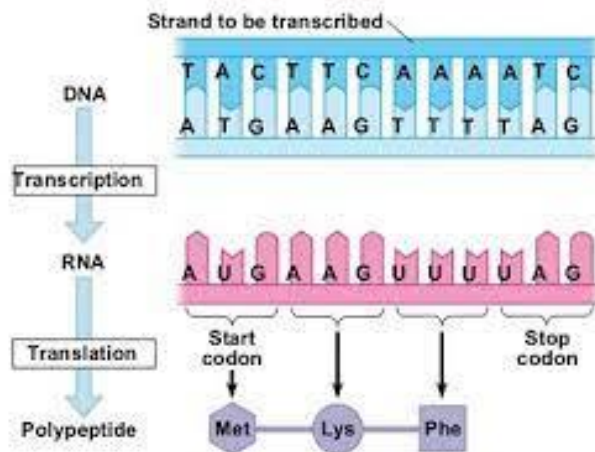
FRAME -1

← GTC

FRAME -2

← TGT

FRAME -3



BOX
5.3

Different 'flavours' of BLAST search different databases

Program	Searches for:	In:
BLASTN	Nucleotide sequence	Nucleotide sequence database
BLASTX	Six-frame translations of a nucleotide sequence	Protein sequence database
BLASTP	Protein sequence	Protein sequence database
TBLASTN	Protein sequence	Six-frame translations of a nucleotide sequence database
TBLASTX	Six-frame translations of a nucleotide sequence	Six-frame translations of a nucleotide sequence database

Step 3: choose a database to search (protein databases)

TABLE 4.1 Protein sequence databases that can be searched by BLAST searching at NCBI. PDB, Protein Data Bank. # indicates approximate number of sequences in database. Adapted from BLAST, NCBI,  <http://blast.ncbi.nlm.nih.gov/>.

Database	Title	# sequences
nr	All nonredundant GenBank CDS translations + PDB + SwissProt + PIR + PRF excluding environmental samples from WGS projects	65 million
Reference proteins	NCBI protein reference sequences	50 million
UniProtKB/SwissProt	Nonredundant UniProtKB/SwissProt sequences	450,000
Patented protein sequences	Protein sequences derived from the Patent division of GenBank	1.3 million
Protein Data Bank	PDB protein database	77,000
Metagenomic proteins	Proteins from WGS metagenomic projects (env_nr)	6.5 million
Transcriptome	Transcriptome Shotgun Assembly (TSA) sequences	770,000

Step 3: choose a database to search (nucleotide)

Database	Title	# sequences
Human Genomic + Transcript	Homo sapiens NCBI Annotation Release 104 RNAs; Homo sapiens all assemblies	55,000
Mouse Genomic + Transcript	Mus musculus NCBI Annotation RNAs; Mus musculus all assemblies	N/A
nr/nt	All GenBank+EMBL+DDBJ+PDB+RefSeq sequences, but excludes EST, STS, GSS, WGS, TSA, patent sequences as well as phase 0, 1, and 2 HTGS sequences	25 million
refseq_rna	NCBI transcript reference sequences	3.5 million
refseq_genomic	NCBI genomic reference sequences	2.7 million
NCBI Genomes	NCBI chromosome sequences	28,000
Expressed sequence tags (EST)	Database of GenBank+EMBL+DDBJ sequences from EST Divisions	75 million
Genomic survey sequences (gss)	Genome survey sequence, includes single-pass genomic data, exon-trapped sequences, and Alu PCR sequences	36 million
High-throughput genomic sequences (HTGS)	Unfinished high-throughput genomic sequences; sequences: phases 0,1 and 2	153,000
Patent sequences	Nucleotide sequences derived from the Patent division of GenBank	21 million
Protein Data Bank	PDB nucleotide database	8000
alu	Human Alu repeat elements	325
Sequence tagged sites (STS)	Database of GenBank+EMBL+DDBJ sequences from STS Divisions	1.3 million
Whole-genome shotgun (wgs)	Whole-genome-shotgun contigs	116 million
Transcriptome Shotgun Assembly (TSA)	Transcriptome shotgun assembly (TSA) sequences	15 million
16S ribosomal RNA sequences (Bacteria and Archaea)	16S ribosomal RNA sequences (bacteria and archaea)	7500

Step 3: choose a database to search

- For BLAST searching, the default option is the **non-redundant** (nr) database
- The nr databases are derived by merging several main protein or DNA databases
 - These databases often contain identical sequences
 - Generally only one of these sequences is retained by the nr database
- The nr databases are often the preferred sites for searching the majority of available sequences

Step 4: optional parameters

You can

- choose the organism to search
- turn filtering on/off
- change the substitution matrix
- change the expect (**e**) value
- change the word size
- change the output format

Step 4: choose optional BLASTP search parameters

The diagram illustrates the mapping of search parameters to the BLASTP interface. On the left, a series of boxes represent the parameters: 'max sequences', 'short queries', 'expect threshold', 'word size', 'max matches', 'scoring matrix', 'gap costs', 'compositional adjustment', 'filter', and 'mask'. Arrows with numbers 1 through 10 point from these boxes to the corresponding fields in the BLASTP interface on the right. The interface is divided into sections: 'Algorithm parameters' (containing 'General Parameters' and 'Scoring Parameters'), and 'Filters and Masking'. The 'BLAST' button and search options are at the bottom.

max sequences → 1 → Max target sequences (100)

short queries → 2 → Short queries (☒ Automatically adjust parameters for short input sequences)

expect threshold → 3 → Expect threshold (10)

word size → 4 → Word size (3)

max matches → 5 → Max matches in a query range (0)

scoring matrix → 6 → Matrix (BLOSUM62)

gap costs → 7 → Gap Costs (Existence: 11 Extension: 1)

compositional adjustment → 8 → Compositional adjustments (Conditional compositional score matrix adjustment)

filter → 9 → Filter (☐ Low complexity regions)

mask → 10 → Mask (☐ Mask for lookup table only, ☐ Mask lower case letters)

BLAST | Search database Non-redundant protein sequences (nr) using ☐ Show results in a new window

Overview of BLAST search strategies

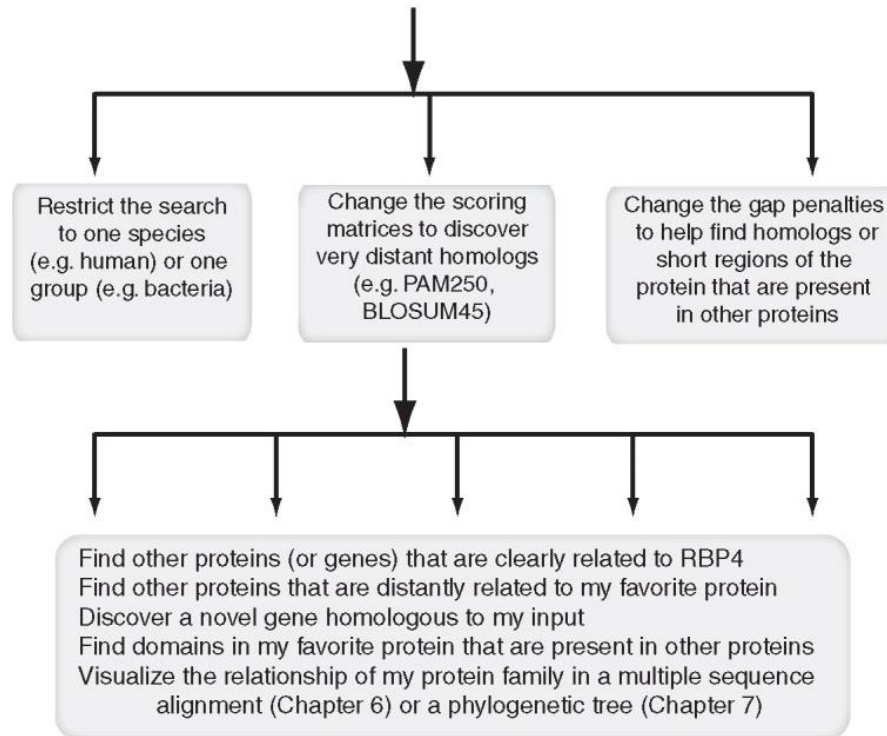
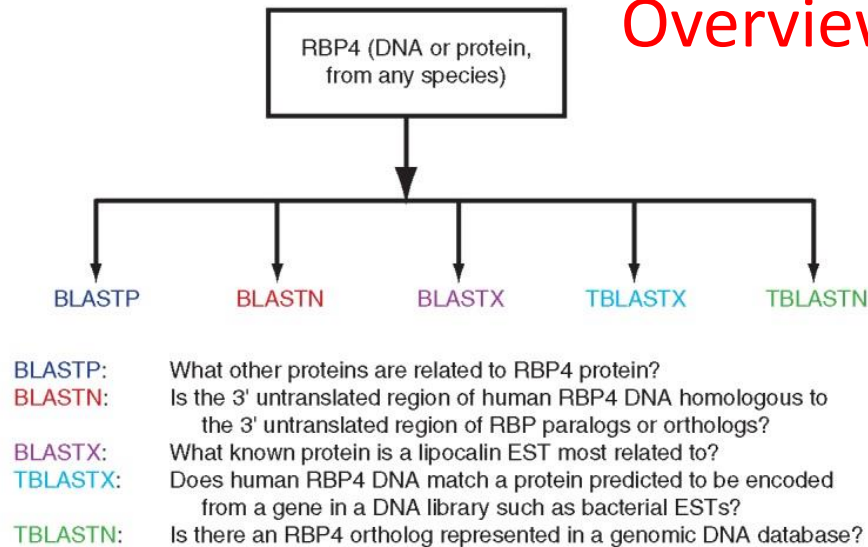
Starting point:
a molecular
sequence

Search
strategies

Sample
questions

Modifiable
search
parameters

Goals:
Results that can
be obtained by
BLAST searching



How to handle too many results

- Select a “**refseq**” database and all the hits that are returned will have RefSeq accession numbers. This will often eliminate redundant database matches
- Limit the database returns by **organism**
- Use just **a portion of the query sequence**, when appropriate
- Adjust the **scoring matrix**
- Adjust the expect value (**E**), lowering E reduces the number of database matches that are returned

How to handle too few results

- Remove Entrez limits,
- Raise the expect values
- Try scoring matrices with higher **PAM** or lower **BLOSUM** value
- Use database-searching algorithms that are more sensitive than BLAST

FASTA search

- FASTA is pronounced “fast A” and stands for “FAST-All” because it is an extension of
 - “FAST-P” for proteins
 - “FAST-N” for nucleotides
- FASTA searches for matching sequence pattern called *k-tuples* (*ktup*) that is similar to the “word” (W) in BLAST
- Default *ktup* is 2 (1-2) for amino acids & 6 (3-6) for nucleotides

FASTA search

Some web-based FASTA servers

- GenomeNet, Japan

<https://www.genome.jp/tools/fasta/>

- EMBL-EBI

<https://www.ebi.ac.uk/Tools/sss/fasta/>

- University of Virginia

https://www.fasta.bioch.virginia.edu/fasta_www2/fasta_list2.shtml

Comparison of BLAST & FASTA

- BLAST begins a search by looking for matches that include exact matches and conservative substitutions; FASTA begins a search by looking at exact matches
- BLAST scans a larger window size than FASTA; hence, FASTA may produce better coverage for homologs
- BLAST may produce multiple best-scoring alignments (also called high-scoring segment pairs or HSPs) from the same sequence; FASTA returns only one alignment from one sequence.

Comparison of BLAST & FASTA

- BLAST automatically masks low-complexity regions; FASTA does not employ such automatic masking. Therefore, if the query sequence has nonunique segments, such as repeats, compositionally biased segments, etc., FASTA search may return alignments with artificially high scores.
- For a given sequence search, the BLAST output is larger than that of FASTA.
- For a given sequence search, BLAST is faster than FASTA.

Outline

BLAST & FASTA

Searching distantly related proteins

Scoring matrix & Statistical significance
of sequence alignment

Position specific iterated BLAST:

PSI-BLAST

- The purpose of **PSI-BLAST** is to look deeper into the database for matches to your query protein sequence by employing a scoring matrix that is customized to your query.
- PSI-BLAST performs more sensitive protein–protein similarity searches is helpful in cases where standard protein–protein BLAST search returns **no significant hits** or comes out with **hypothetical proteins**. PSI-BLAST can be used for **Superfamily** and **Functional** analysis

The algorithm for PSI-BLAST

- PSI-BLAST makes a standard protein–protein BLAST search in the first round and generates a multiple alignment of the **highest scoring pairs** (HSP)
- Calculates a profile or a **position-specific score matrix** (PSSM) from the multiple alignment of related proteins
- Newly generated profile will be used for searching
- If **new sequences** are detected above the mentioned **e-value cut-off (threshold)**, it will be again added to alignment and the profile is reconstructed for another round of searching

The algorithm for PSI-BLAST

- This process continues iteratively until desired or no new sequences are detected above the defined threshold (which is called convergence)
- The process makes PSI-BLAST far more effective in detecting **distantly related proteins** than BLASTP which performs a single query alone

Protein BLAST: search protein d X +

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Getting Started SBT WORKING SCHED... New Bookmark Welcome, Nguyễn - B... Trường Đại học Quốc ... Trường Đại học Quốc ... THỰC ĐƠN - Trường ...

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. [more...](#) [Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

From

To

Or, upload file [Browse...](#) No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database [?](#)

Organism [Optional](#) ☐ exclude [+](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude [Optional](#) ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

- ☐ Quick BLASTP (Accelerated protein-protein BLAST)
- ☐ blastp (protein-protein BLAST)
- ☒ PSI-BLAST (Position-Specific Iterated BLAST)
- ☐ PHI-BLAST (Pattern Hit Initiated BLAST)
- ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST Search database nr using PSI-BLAST (Position-Specific Iterated BLAST)

☐ Show results in a new window

Type here to search

ENG INTL 08

PSI-BLAST for accession number: ADM97672

Protein BLAST: search protein d x +

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Getting Started SBT WORKING SCHED... New Bookmark Welcome, Nguyễn - B... Trường Đại học Quốc ... Trường Đại học Quốc ... THỰC ĐƠN - Trường ...

Scoring Parameters

Matrix BLOSUM62 ⓘ

Gap Costs Existence: 11 Extension: 1 ⓘ

Compositional adjustments Conditional compositional score matrix adjustment ⓘ

Filters and Masking

Filter ☐ Low complexity regions ⓘ

Mask ☐ Mask for lookup table only ⓘ
☐ Mask lower case letters ⓘ

PSI/PHI/DELTA BLAST

Upload PSSM Optional Browse... No file selected. ⓘ

PSI-BLAST Threshold 0.005 ⓘ

Pseudocount 0 ⓘ

BLAST Search database nr using PSI-BLAST (Position-Specific Iterated BLAST)
☐ Show results in a new window

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NCBI

National Center for Biotechnology Information, U.S. National Library of Medicine
8600 Rockville Pike, Bethesda MD, 20894 USA

[Policies and Guidelines](#) | [Contact](#)

[Support](#)



Type here to search



Pattern-Hit Initiated (PHI)-BLAST

- **PHI-BLAST** is yet another version of BLAST that not only looks for similar sequences but also a **signature**
- Protein sequence analyses have often observed that a group of related proteins always had '**elements**' which are conserved patterns of amino acids known as '**motifs**' or '**signatures**'
- PHI-BLAST takes only **patterns of domains** and **protein families** that have been studied and characterized by **Prosite**

<https://prosite.expasy.org/>



Database of protein domains, families and functional sites



SARS-CoV-2 relevant PROSITE motifs

PROSITE consists of documentation entries describing protein domains, families and functional sites as well as associated patterns and profiles to identify them [\[More... / References / Commercial users\]](#).

PROSITE is complemented by [ProRule](#), a collection of rules based on profiles and patterns, which increases the discriminatory power of profiles and patterns by providing additional information about functionally and/or structurally critical amino acids [\[More...\]](#).

Release 2020_05 of 07-Oct-2020 contains 1865 documentation entries, 1311 patterns, 1288 profiles and 1321 ProRule.

Search

 e.g. PDOC00022, PS50089, SH3, zinc finger

Browse

- [by documentation entry](#)
- [by ProRule description](#)
- [by taxonomic scope](#)
- [by number of positive hits](#)

Quick Scan mode of ScanProsite

Quickly find matches of your protein sequences to PROSITE signatures (max. 10 sequences). [\[?\]](#) [Examples](#)

Enter UniProtKB accessions or identifiers or PDB identifiers or sequences in FASTA format

Other tools

- [PRATT](#) - allows to interactively generate conserved patterns from a series of unaligned proteins.
- [MyDomains - Image Creator](#) - allows to generate custom domain figures.



Protein BLAST: search protein d x +

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Getting Started SBT WORKING SCHED... New Bookmark Welcome, Nguyễn - B... Trường Đại học Quốc ... Trường Đại học Quốc ... THỰC ĐƠN - Trường ...

Click [here](#) for more info.

Or, upload file
Browse... No file selected.

Job Title
Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database
Non-redundant protein sequences (nr)

Organism
Optional
Enter organism name or id—completions will be suggested
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Exclude
Optional
☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

- ☐ Quick BLASTP (Accelerated protein-protein BLAST)
- ☐ blastp (protein-protein BLAST)
- ☐ PSI-BLAST (Position-Specific Iterated BLAST)
- ☒ PHI-BLAST (Pattern Hit Initiated BLAST)
Enter a PHI pattern
- ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)
Choose a BLAST algorithm

BLAST
Search database nr using PHI-BLAST (Pattern Hit Initiated BLAST)
☐ Show results in a new window

Algorithm parameters

General Parameters

Max target sequences
500
Select the maximum number of aligned sequences to display

Short queries
☒ Automatically adjust parameters for short input sequences

PHI-BLAST for accession number: P23825

Protein BLAST: search protein d X +

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Getting Started X Lịch tuần các đơn vị.xl... Blackboard Learn

Or, upload file Browse... P23825.fasta

Job Title

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database Protein Data Bank proteins(pdb)

Organism Optional

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Exclude Optional

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☒ PHI-BLAST (Pattern Hit Initiated BLAST)

T]-x(3,4)-[GNEP]-x(3,6)-C-[NES]-[ASNR]-C

Enter a PHI pattern

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST

Search database pdb using PHI-BLAST (Pattern Hit Initiated BLAST)

☐ Show results in a new window

+ Algorithm parameters

Note: Parameter values that differ from the default are highlighted in yellow and marked with ♦ sign

PHI-BLAST for accession number: P23825

Search

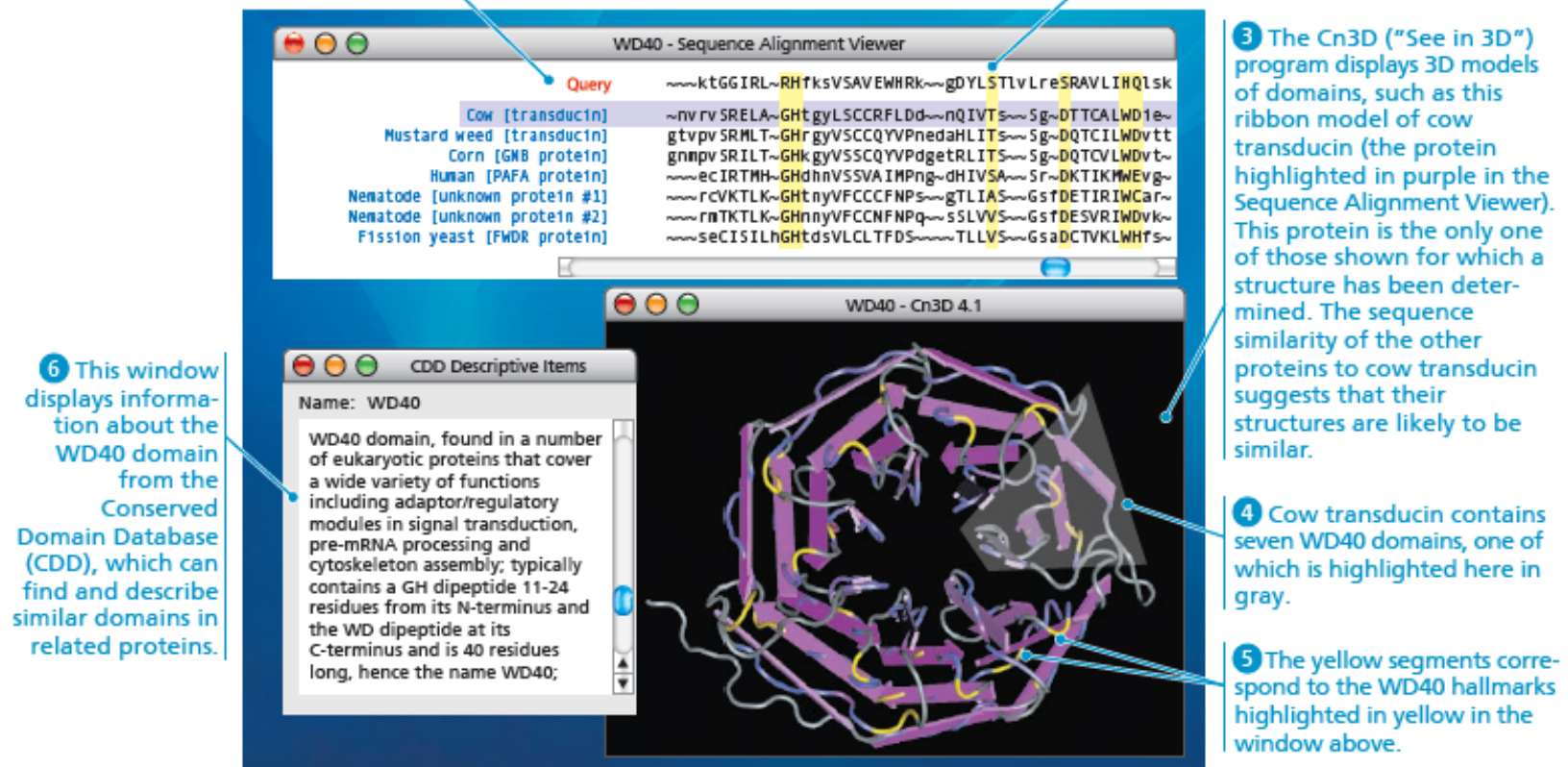
GATA-binding factor 3

The GATA families of transcription factors are proteins that bind to DNA sites having consensus sequence (A/T)GATA(A/G) that is found within the regulatory region of a number of genes. GATA-3 is a transcriptional activator that binds to the enhancer of the T-cell receptor α and delta genes

▼ **Figure 21.3 National Center for Biotechnology Information (NCBI) website.** Some results are shown from a search for regions of proteins similar to an amino acid sequence in a muskmelon protein.

1 In this window, a partial amino acid sequence from an unknown muskmelon protein ("Query") is aligned with sequences from other proteins that the program found to be similar. Each sequence represents a domain called WD40.

2 Four hallmarks of the WD40 domain are highlighted in yellow. (Sequence similarity is based on chemical aspects of the amino acids, so the amino acids in each hallmark region are not always identical.)



Domain Enhanced Lookup Time Accelerated-BLAST (DELTA-BLAST)

- In **PSI-BLAST**, the program automatically generates multiple sequence alignments and their related PSSMs from the results of database searches, iteratively
- **DELTA-BLAST** searches a database containing pre-constructed PSSMs generated from **Conserved Domain Database** (CDD),
 - CDD is an NCBI resource that identifies conserved domains within protein sequences, before making a search in a protein sequence database

Domain Enhanced Lookup Time Accelerated-BLAST (DELTA-BLAST)

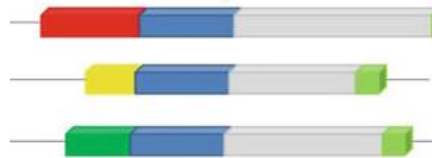
- DELTA-BLAST initially uses **RPS-BLAST** (Reverse PSI-BLAST) to align a query sequence to conserved domains in CDD
- Later performs a sequence database search using a PSSM derived from the aligned domains

Query

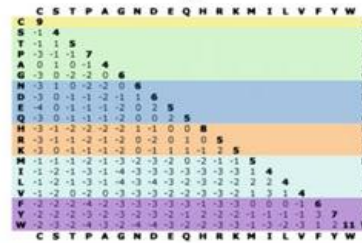
LWMWMYPKSCGLQPEECMCH



CDD Searching



Multiple alignment of Conserved Domain



Generate PSSM



Search Sequence Database

Protein BLAST: search protein

https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Enter accession number(s), gi(s), or FASTA sequence(s)

Query subrange

From

To

Or, upload file

Browse... No file selected.

Job Title

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database

Reference proteins (refseq_protein)

Organism

Optional

Arabisidopsis thaliana (taxid:3702) ☐ exclude +

Nostoc PCC 7524 (taxid:28072) ☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Exclude

Optional

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☒ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

BLAST

Search database refseq_protein using DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

☐ Show results in a new window

Algorithm parameters

Note: Parameter values that differ from the default are highlighted in yellow and marked with ♦ sign

Restore default search parameters

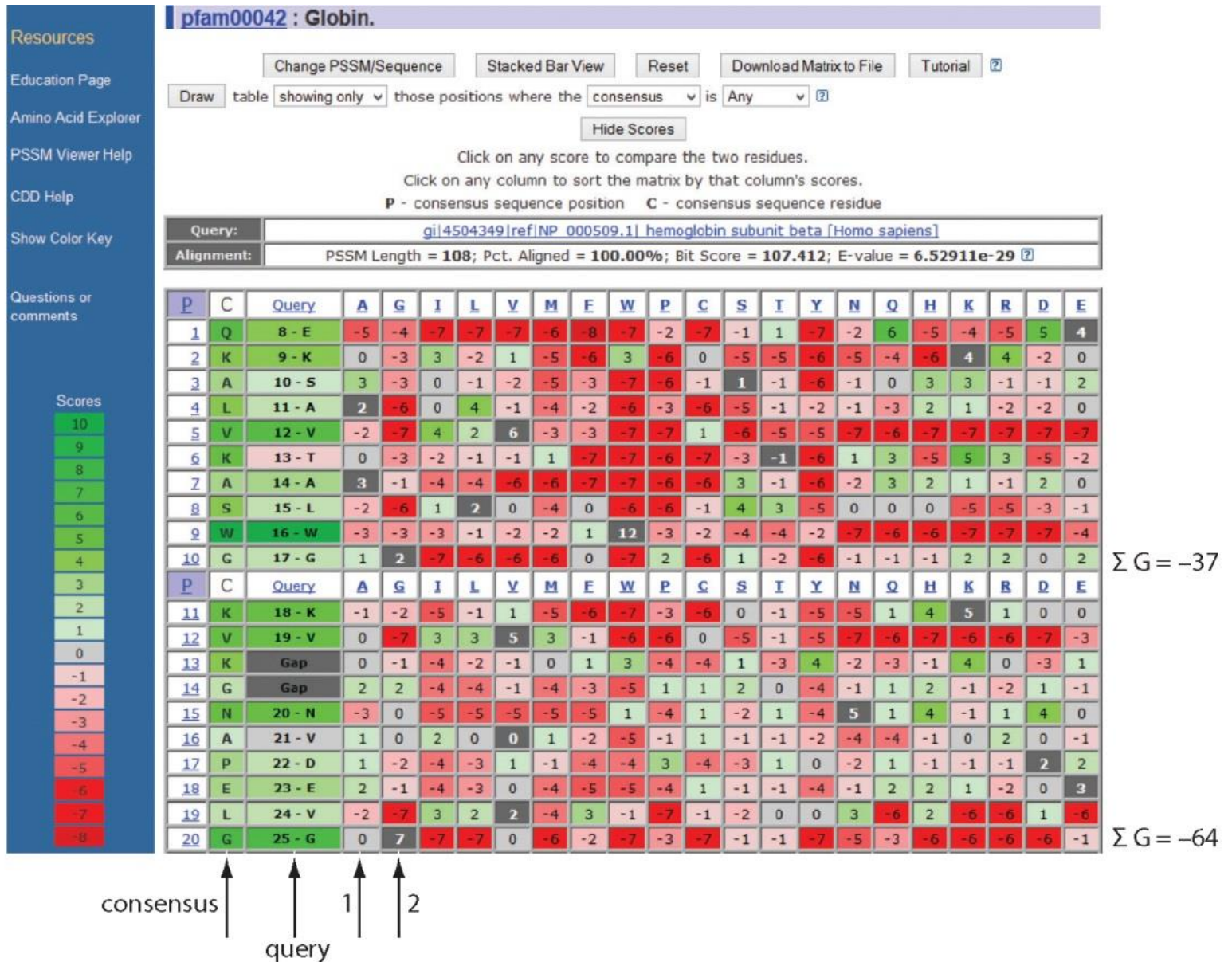
General Parameters

Max target

500

DELTA-BLAST for accession number: CAA60423

Position-specific scoring matrix (PSSM)



Amino acid	Three-letter code	One-letter code	Comment
Glycine	Gly	G	Only -H as side chain Side chain to N bond
Alanine	Ala	A	
Valine	Val	V	
Leucine	Leu	L	
Isoleucine	Ile	I	
Proline	Pro	P	
Phenylalanine	Phe	F	
Methionine	Met	M	
Tryptophan	Trp	W	
Cysteine	Cys	C	Forms disulfide bonds
Asparagine	Asn	N	Amide N polar
Glutamine	Gln	Q	Amide N polar
Serine	Ser	S	-OH group polar
Threonine	Thr	T	-OH group polar
Tyrosine	Tyr	Y	-OH group polar
Aspartic acid	Asp	D	
Glutamic acid	Glu	E	
Histidine	His	H	
Lysine	Lys	K	
Arginine	Arg	R	

20 amino acids
colored according to
their properties

	Nonpolar
	Polar
	Acidic
	Basic

Inspect the BLASTP output to identify empirical “rules” regarding amino acids tolerated at each position

730496	66	FTVDENGQMSATAKGRVRLFNWWDVCADMIGSFTDTEPAKFKMKYWGVASFLQKGNDH	125
200679	63	FSVDEKGHMSATAKGRVRLLSNWEVCADMVGTFTDTEPAKFKMKYWGVASFLQKGNDH	122
206589	34	FSVDEKGHMSATAKGRVRLLSNWEVCADMVGTFTDTEPAKFKMKYWGVASFLQKGNDH	93
2136812	2	MSATAKGRVRLLSNWWDVCADMVGTFTDTEPAKFKMKYWGVASFLQKGNDH	53
132408	65	FKIEDNGKTTATAKGRVRILDKLELCANMVGTFTETNDPAKYRMKYHGALAILERGLDDH	124
267584	44	FSVDESGKVTATAHGRVILNNWEMCANMFGTFEDTPDPAKFKMRYWGAAASYLQTGNDH	103
267585	44	FSVDGSGKVTATAQGRVILNNWEMCANMFGTFEDTPDPAKFKMRYWGAAASYLQSGNDH	103
8777608	63	FTIHEDGAMTATAKGRVILNNWEMCADMMATFETTPDPAKFRMRYWGAAASYLQTGNDH	122
6687453	60	FKVEEDGTMTATAIGRVILNNWEMCANMFGTFEDTEPAKFKMKYWGAAASYLQGYDDH	119
10697027	81	FKVQEDGTMTATATGRVILNNWEMCANMFGTFEDTEEPARFKMRYWGAAASYLQGYDDH	140
13645517	1	MVGTFTDTEPAKFKMKYWGVASFLQKGNDH	32
13925316	38	FSVDGSGKMTATAQGRVILNNWEMCANMFGTFEDTPDPAKFKMRYWGAAASYLQSGNDH	97
131649	65	YTVEEDGTMTASSKGRVKLFGFWVICADMAAQYTDPTTPAKMYMTYQGLASYLSSGGDNY	126

R,I,K

C

D,E,T

K,R,T

N,L,Y,G

		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
1	M	-1	-3	-3	-3	-3	-1	-3	-3	-3	-1	-3	-3	-3	-3	-3	-3	-1	-3	-1	1
2	K	-1	1	0	1	-4	2	4	-2	0	-3	-3	3	-2	-4	-1	0	-1	-3	-2	-3
3	W	-3	-3	-4	-5	-3	-2	-3	-3						1	-4	-3	-3	12	2	-3
4	V	0	-3	-3	-4	-1	-3	-3	-4						1	-3	-2	0	-3	-1	4
5	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
6	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
7	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
8	L	-1	-3	-3	-4	-1	-3	-3	-4	-3	2	2	-3	1	3	-3	-2	-1	-2	0	3
9	L	-1	-3	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	2
10	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
11	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
12	A	5																			
13	W	-2																			
14	A	3																			
15	A	2																			
16	A	4																			
...																					
37	S	2	-1	0	-1	-1	0	0	0	-1	-2	-3	0	-2	-3	-1	4	1	-3	-2	-2
38	G	0	-3	-1	-2	-3	-2	-2	6	-2	-4	-4	-2	-3	-4	-2	0	-2	-3	-3	-4
39	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-3	-2	0
40	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
41	Y	-2	-2	-2	-3	-3	-2	-2	-3	2	-2	-1	-2	-1	3	-3	-2	-2	2	7	-1
42	A	4	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0

20 amino acids

all the amino acids
from position 1 to the
end of your PSI-BLAST
query protein

		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
1	M	-1	-2	-2	-3	-2	-1	-2	-3	-2	1	2	-2	6	0	-3	-2	-1	-2	-1	1
2	K	-1	1	0	1	-4	2	4	-2	0	-3	-3	3	-2	-4	-1	0	-1	-3	-2	-3
3	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
4	V	0	-3	-3	-4	-1	-3	-3	-4	-4	3	1	-3	1	-1	-3	-2	0	-3	-1	4
5	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
6	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
7	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
8	L	-1	-3	-3	-4	-1	-3	-3	-4	-3	2	2	-3	1	3	-3	-2	-1	-2	0	3
9	L	-1	-3	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	2
10	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
11	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
12	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
13	W	-2	-3	-4	-4	-2	-2	-3	-4	-3	1	4	-3	2	1	-3	-3	-2	7	0	0
14	A	3	-2	-1	-2	-1	-1	-2	4	-2	-2	-2	-1	-2	-3	-1	1	-1	-3	-3	-1
15	A	2	-1	0	-1	-2	2	0	2	-1	-3	-3	0	-2	-3	-1	3	0	-3	-2	-2
16	A	4	-2	-1	-2	-1	-1	-1	3	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	-1
...																					
37	S	2	-1	0	-1	-1	0	0	0	-1	-2	-3	0	-2	-3	-1	4	1	-3	-2	-2
38	G	0	-3	-1	-2	-3	-2	-2	6	-2	-4	-4	-2	-3	-4	-2	0	-2	-3	-3	-4
39	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-3	-2	0
40	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
41	Y	-2	-2	-2	-3	-3	-2	-2	-3	2	-2	-1	-2	-1	3	-3	-2	-2	2	7	-1
42	A	4	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0

		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
1	M	-1	-2	-2	-3	-2	-1	-2	-3	-2	1	2	-2	6	0	-3	-2	-1	-2	-1	1
2	K	-1	1	0	1	-4	2	4	-2	0	-3	-3	3	-2	-4	-1	0	-1	-3	-2	-3
3	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
4	V	0	-3	-3	-4	-1	-3	-3	-4	-4	3	1	-3	1	-1	-3	-2	0	-3	-1	4
5	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
6	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
7	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
8	L	-1	-3	-3	-4	-1	-3	-3	-4	-3	2	2	-3	1	3	-3	-2	-1	-2	0	3
9	L	-1	-3	-4	-4										0	-3	-3	-1	-2	-1	2
10	L	-2	-2	-4	-4										0	-3	-3	-1	-2	-1	1
11	A	5	-2	-2	-2										-3	-1	1	0	-3	-2	0
12	A	5	-2	-2	-2										-3	-1	1	0	-3	-2	0
13	W	-2	-3	-4	-4										1	-3	-3	-2	7	0	0
14	A	3	-2	-1	-2										-3	-1	1	-1	-3	-3	-1
15	A	2	-1	0	-2										-3	-1	3	0	-3	-2	-2
16	A	4	-2	-1	-2										-3	-1	1	0	-3	-2	-1
...																					
37	S	2	-1	0	-1										-3	-1	4	1	-3	-2	-2
38	G	0	-3	-1	-2										-4	-2	0	-2	-3	-3	-4
39	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-3	-2	0
40	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
41	Y	-2	-2	-2	-3	-3	-2	-2	-3	2	-2	-1	-2	-1	3	-3	-2	-2	2	7	-1
42	A	4	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0

note that a given amino acid (such as alanine) in your query protein can receive different scores for matching alanine—depending on the position in the protein

		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
1	M	-1	-2	-2	-3	-2	-1	-2	-3	-2	1	2	-2	6	0	-3	-2	-1	-2	-1	1
2	K	-1	1	0	1	-4	2	4	-2	0	-3	-3	3	-2	-4	-1	0	-1	-3	-2	-3
3	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
4	V	0	-3	-3	-4	-1	-3	-3	-4	-4	3	1	-3	1	-1	-3	-2	0	-3	-1	4
5	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
6	A	5	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0
7	L	-2	-2	-4	-4	-1	-2	-3	-4	-3	2	4	-3	2	0	-3	-3	-1	-2	-1	1
8	L	-1	-3	-3	-4										3	-3	-2	-1	-2	0	3
9	L	-1	-3	-4	-4										0	-3	-3	-1	-2	-1	2
10	L	-2	-2	-4	-4										0	-3	-3	-1	-2	-1	1
11	A	5	-2	-2	-2										-3	-1	1	0	-3	-2	0
12	A	5	-2	-2	-2										-3	-1	1	0	-3	-2	0
13	W	-2	-3	-4	-4												-3	-2	7	0	0
14	A	3	-2	-1	-2										-3	-1	1	-1	-3	-3	-1
15	A	2	-1	0	-1										-3	-1	3	0	-3	-2	-2
16	A	4	-2	-1	-2										-3	-1	1	0	-3	-2	-1
...																					
37	S	2	-1	0	-1										-3	-1	4	1	-3	-2	-2
38	G	0	-3	-1	-2										-3	-1	4	1	-3	-2	-2
39	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-3	-2	0
40	W	-3	-3	-4	-5	-3	-2	-3	-3	-3	-3	-2	-3	-2	1	-4	-3	-3	12	2	-3
41	Y	-2	-2	-2	-3	-3	-2	-2	-3	2	-2	-1	-2	-1	3	-3	-2	-2	2	7	-1
42	A	4	-2	-2	-2	-1	-1	-1	0	-2	-2	-2	-1	-1	-3	-1	1	0	-3	-2	0

note that a given amino acid (such as tryptophan) in your query protein can receive different scores for matching tryptophan—depending on the position in the protein

Outline

BLAST & FASTA

Searching distantly related proteins

Scoring matrix & Statistical significance
of sequence alignment

Scoring matrix

Alignment score:

$$S = \Sigma_i + \Sigma_m - G_t$$

- S = raw score
- Σ_i = total score for identities
- Σ_m = total score for mismatches
- G_t = total gap penalty

Query: a t t a g c a c t g c a t t a g g a t c a c g

| | | | | | | | | | | | | | | | | | | | |

Subject: a t a t g c a c t g c a t t a g g c t c a c c

11 bp word

←—————→

+1 +1	+1 +1 +1 +1 +1 +1 +1 +1 +1 +1 +1 +1 +1 +1	+1 +1 +1 +1	= +19
-3 -3		-3 -3	= -12

*Total = +7

- * The "Total" represents a raw score. BLAST uses statistics to convert the raw score to a bit score. For details, please see Altschul SF, et. al. (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.* 25(17):3389-402.

How raw scores are calculated: an example

Score = 18.1 bits (35), Expect = 0.015, Method: Composition-based stats.
Identities = 11/24 (45%), Positives = 12/24 (50%), Gaps = 2/24 (8%)

```
Query 12 VTALWGKVNVD--EVGGEALGRLL 33
          V +WGKV D G E L RL
Sbjct 11 VLNWVGKVEADIPGHGQEV LIRLF 34

match      4 11 5 6 6 5 4 5 sum of matches: +60 (round up to +61)
           6 4 4
mismatch   -1 1 0 -2 -2 -4 0 sum of mismatches: -13
           -2 0 -3 0
gap open                -11 sum of gap penalties: -13
gap extend              -2

total raw score: 61 - 13 - 13 = 35
```

For a set of aligned residues we assign scores based on matches, mismatches, gap open penalties, and gap extension penalties. These scores add up to the total raw score.

Scoring matrix

- For both nucleic acids and proteins, the alignment score is calculated using a scoring matrix

A scoring matrix is a set of values representing the likelihood of one residue being substituted by another during sequence divergence through evolution

- Scoring matrix is also known as the **substitution matrix**
- Scoring matrices for nucleic acids: **NUC4.2** & **NUC4.4DNA**
- Scoring matrices for proteins: **PAM** & **BLOSUM**

Substitution Matrix

- A substitution matrix contains values proportional to the probability that amino acid i mutates into amino acid j for all pairs of amino acids
- Substitution matrices are constructed by assembling a large and diverse sample of verified pairwise alignments (or multiple sequence alignments) of amino acids
- Substitution matrices should reflect the true probabilities of mutations occurring through a period of evolution

PAM matrix

- **PAM** – **P**oint **A**ccepted **M**utation, based on observed substitution frequencies between closely related homologous proteins, varies with the number of accepted point mutations per 100 residues
- PAM matrices are based on global alignments of closely related proteins
- The **PAM1** is the matrix calculated from comparisons of sequences with no more than 1% divergence. At an evolutionary interval of PAM1, **one change** has occurred over a length of **100 amino acids**.

PAM matrix

- The suffix (**number**) with PAM indicates evolutionary distance
 - Higher PAM number is better in aligning more divergent sequences, eg. PAM250 vs. PAM120
 - Lower suffix number is better in dealing with evolutionary closer protein alignment

PAM120

C	9																			
S	-1	3																		
T	-3	2	4																	
P	-3	1	-1	6																
A	-3	1	1	1	3															
G	-5	1	-1	-2	1	5														
N	-5	1	0	-2	0	0	4													
D	-7	0	-1	-2	0	0	2	5												
E	-7	-1	-2	-1	0	-1	1	3	5											
Q	-7	-2	-2	0	-1	-3	0	1	2	6										
H	-4	-2	-3	-1	-3	-4	2	0	-1	3	7									
R	-4	-1	-2	-1	-3	-4	-1	-3	-3	1	1	6								
K	-7	-1	-1	-2	-2	-3	1	-1	-1	0	-2	2	5							
M	-6	-2	-1	-3	-2	-4	-3	-4	-4	-1	-4	-1	0	8						
I	-3	-2	0	-3	-1	-4	-2	-3	-3	-3	-4	-2	-2	1	6					
L	-7	-4	-3	-3	-3	-5	-4	-5	-4	-2	-3	-4	-4	3	1	5				
V	-2	-2	0	-2	0	-2	-3	-3	-3	-3	-3	-3	-4	1	3	1	5			
F	-6	-3	-4	-5	-4	-5	-4	-7	-6	-6	-2	-4	-6	-1	0	0	-3	8		
Y	-1	-3	-3	-6	-4	-6	-2	-5	-4	-5	-1	-6	-6	-4	-2	-3	-3	4	8	
W	-8	-2	-6	-7	-7	-8	-5	-8	-8	-6	-5	1	-5	-7	-7	-5	-8	-1	-1	12
	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W

Color shading indicates different physico-chemical properties of amino acids

Correspondence between observed differences (per 100 residues) and evolutionary distance (in PAMs)

Observed differences in 100 residues	Evolutionary distance in PAMs	
1	1.0	
5	5.1	
10	10.7	
15	16.6	
20	23.1	
25	30.2	
30	38.0	
35	47	
40	56	
45	67	
50	80	PAM 80
55	94	
60	112	
65	133	
70	159	
75	195	
80	246	PAM 250

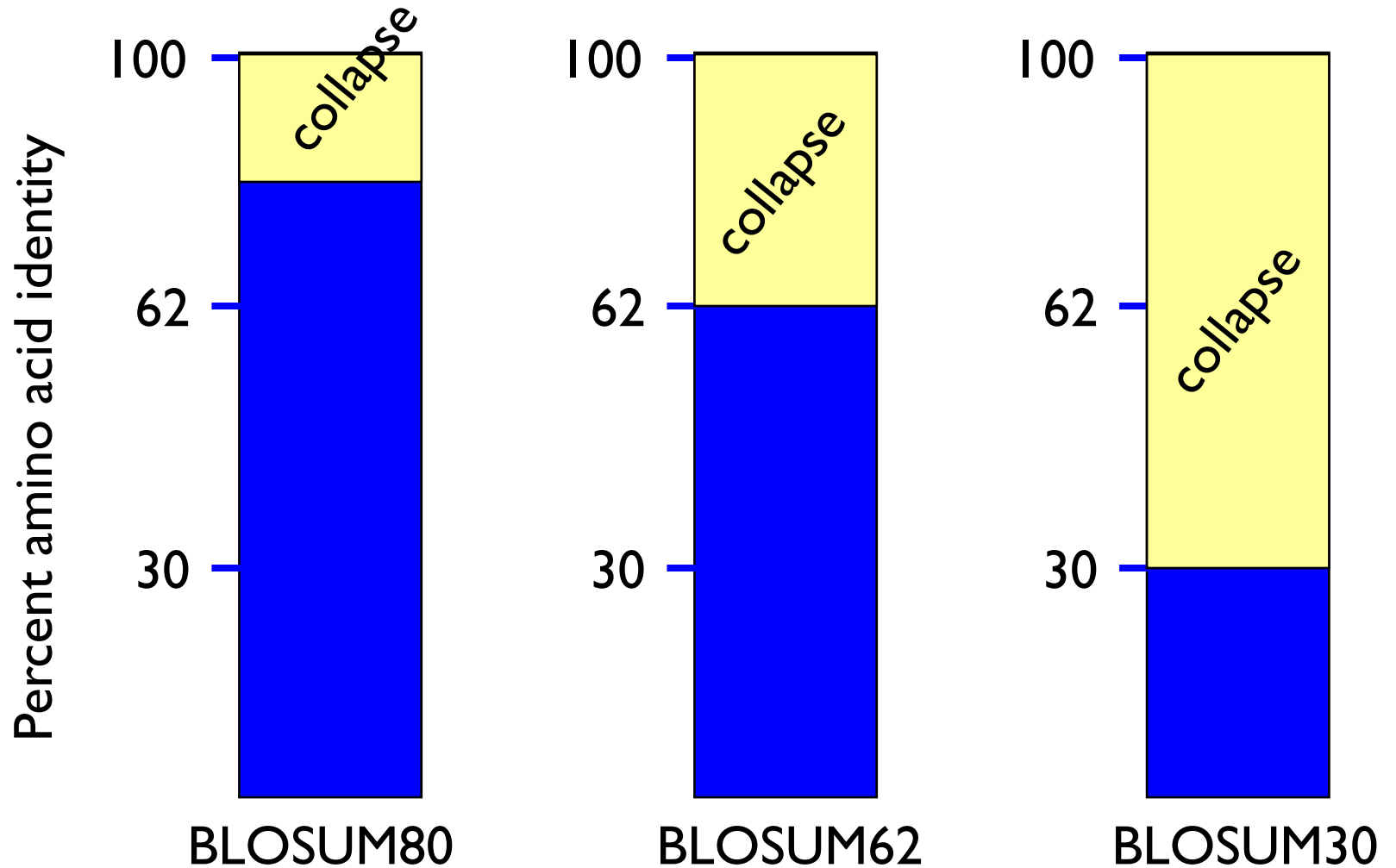
BLOSUM matrix

- **BLOSUM** – **BLO**ck **SU**bstitution **M**atrix, based on substitution data from highly conserved regions of homologous proteins, varies with cut-off % identity used in calculation, eg. BLOSUM50, BLOSUM80
 - **BLOSUM62** is a matrix calculated from comparisons of sequences with **no less than 62% divergence**
- BLOSUM matrices are based on local alignments
- For BLOSUM matrices, the suffix numbering system is the opposite of PAM matrices
 - Higher suffix number is better in dealing with evolutionary closer protein alignment

BLOSUM62

C	9																			
S	-1	4																		
T	-1	1	5																	
P	-3	-1	-1	7																
A	0	1	0	-1	4															
G	-3	0	-2	-2	0	6														
N	-3	1	0	-2	-2	0	6													
D	-3	0	-1	-1	-2	-1	1	6												
E	-4	0	-1	-1	-1	-2	0	2	5											
Q	-3	0	-1	-1	-1	-2	0	0	2	5										
H	-3	-1	-2	-2	-2	-2	1	-1	0	0	8									
R	-3	-1	-1	-2	-1	-2	0	-2	0	1	0	5								
K	-3	0	-1	-1	-1	-2	0	-1	1	1	-1	2	5							
M	-1	-1	-1	-2	-1	-3	-2	-3	-2	0	-2	-1	-1	5						
I	-1	-2	-1	-3	-1	-4	-3	-3	-3	-3	-3	-3	-3	1	4					
L	-1	-2	-1	-3	-1	-4	-3	-4	-3	-2	-3	-2	-2	2	2	4				
V	-1	-2	0	-2	0	-3	-3	-3	-2	-2	-3	3	2	1	3	1	4			
F	-2	-2	-2	-4	-2	-3	-3	-3	-3	-3	-1	-3	-3	0	0	0	-1	6		
Y	-2	-2	-2	-3	-2	-3	-2	-3	-2	-1	2	-2	-2	-1	-1	-1	-1	3	7	
W	-2	-3	-2	-4	-3	-2	-4	-4	-3	-2	-2	-3	-3	-1	-3	-2	-3	1	2	11
	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W

BLOSUM Matrices



PAM or BLOSUM

- All BLOSUM matrices perform better than PAM matrices
- For less divergent proteins: PAM100 $\approx$ BLOSUM90
- For most other proteins:
PAM120 $\approx$ BLOSUM80
PAM160 $\approx$ BLOSUM62
PAM200 $\approx$ BLOSUM52
- For more divergent proteins: PAM250 $\approx$ BLOSUM45

Summary of PAM and BLOSUM matrices

BLOSUM90

BLOSUM62

BLOSUM45

PAM30

PAM120

PAM250

Less divergent



More divergent

Human versus
chimpanzee beta globin

Human versus
bacterial globins

A higher PAM number, and a lower BLOSUM number, tends to correspond to a matrix tuned to more divergent proteins.

Statistical significance of sequence alignment

- Different alignments should not be directly compared based on their raw score (**S**)
- Different alignments should only be compared after **normalization**. This is achieved by determining the **statistical significance** of the score.
- **P-value**: probability of obtaining a score $\geq S$ by chance
 - The P-value ranges from **0** to **1**; the closer it is to 0, the better is the alignment.

Normalization of raw score

- **Bit score** (S') is a normalized raw score expressed in bits
- Usually, good alignments produce a bit score **> 50**
- Bit scores are comparable between different searches because they are normalized to account for the use of different scoring matrices and different database sizes

Statistical significance of sequence alignment

- **E-value** is the **expectation value** that indicates the number of alignments with a score $\geq S$ that one can expect to find by chance in a database of size N
 - The E-value is dependent on the database size and the query length
- For BLAST (both nucleotide and protein), the **default E-value** is set at **0.05** in the Expect threshold box under Algorithm parameters
- The default E-value should be increased if searching for short sequence matches, because setting a lower E-value will automatically exclude the short matches as spurious and these will not be reported

Statistical **p-value** ?

$$1\text{e-}4 = 0.0001$$

Statistical significance of sequence alignment

- The typical threshold for the E-value when judging homology, particularly using BLAST, is $E \leq 1e-5$ ($=1 \times 10^{-5}$)
 - The lower the E-value, the more significant the match is
- For **nucleic-acid-based** search, the suggested threshold (minimum significant hit) for the E-value is $\leq 1e-6$ ($=10^{-6}$), and a sequence identity of $\geq 70\%$
- For protein-based search, the suggested threshold for the E-value $\leq 1e-4$ ($=10^{-4}$), with a sequence identity of $\geq 35\%$

How to interpret BLAST: E values and p values

E values of about 1 to 10 are far easier to interpret than corresponding p values.

Very small E values are very similar to p values.

<u>E</u>	<u>p</u>
10	0.99995460
5	0.99326205
2	0.86466472
1	0.63212056
0.1	0.09516258 (about 0.1)
0.05	0.04877058 (about 0.05)
0.001	0.00099950 (about 0.001)
0.0001	0.00010000

E values are comparable to p values and are designed to be more convenient to interpret.

Statistical significance of sequence alignment

To summarize the utility of the statistical estimates of sequence alignment in simple terms, **the better the alignment** (e.g. homologous sequences), **the lower the P- and E-values**, and **the higher bit score**

[Edit Search](#) [Save Search](#) [Search Summary](#) [How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

Job Title ref[NM_006744.3]
RID [NMUZ2JW4013](#) *Search expires on 10-05 10:50 am* [Download All](#) [▼](#)
Program BLASTN [Citation](#) [▼](#)
Database nt [See details](#) [▼](#)
Query ID [NM_006744.3](#)
Description Homo sapiens retinol binding protein 4 (RBP4), transcript ...
Molecule type rna
Query Length 941
Other reports [Distance tree of results](#) [MSA viewer](#) [?](#)

Filter Results

Organism *only top 20 will appear* ☐ exclude
Type common name, binomial, taxid or group name
[+ Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

[Filter](#) [Reset](#)

[Descriptions](#) [Graphic Summary](#) [Alignments](#) [Taxonomy](#)

Sequences producing significant alignments [Download](#) [New](#) [Select columns](#) [Show](#) [?](#)

☒ select all 100 sequences selected

[GenBank](#) [Graphics](#) [Distance tree of results](#) [New](#) [MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Homo sapiens retinol binding protein 4 (RBP4), transcript variant 1, mRNA	Homo sapiens	1674	1674	98%	0.0	100.00%	1070	NM_006744.4
☒	PREDICTED: Gorilla gorilla gorilla retinol binding protein 4 (RBP4), mRNA	Gorilla gorilla go...	1656	1656	98%	0.0	99.57%	953	XM_019035438.2
☒	Homo sapiens retinol binding protein 4, plasma, mRNA (cDNA clone MGC:22269 IMAGE:4767481), complete cds	Homo sapiens	1656	1656	100%	0.0	99.04%	950	BC020633.1
☒	PREDICTED: Pan troglodytes retinol binding protein 4 (RBP4), transcript variant X1, mRNA	Pan troglodytes	1645	1645	98%	0.0	99.25%	967	XM_009458876.3
☒	PREDICTED: Pan paniscus retinol binding protein 4 (RBP4), transcript variant X1, mRNA	Pan paniscus	1631	1631	98%	0.0	98.92%	990	XM_034931118.1
☒	PREDICTED: Nomascus leucogenys retinol binding protein 4 (RBP4), transcript variant X1, mRNA	Nomascus leuc...	1613	1613	98%	0.0	98.49%	986	XM_003255233.3
☒	PREDICTED: Hylobates moloch retinol binding protein 4 (RBP4), transcript variant X2, mRNA	Hylobates moloch	1607	1607	98%	0.0	98.39%	985	XM_032136583.1

Other reports ?

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

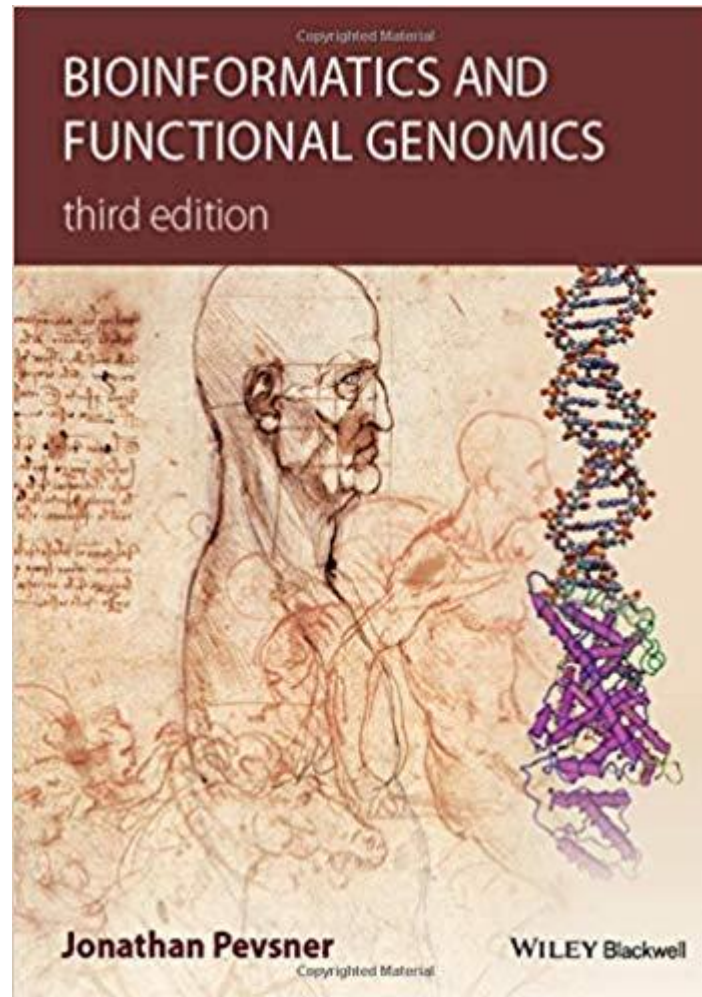
Sequences producing significant alignments Download New Select columns Show 100 ?

☒ select all 100 sequences selected

GenPept Graphics

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Mamu class II histocompatibility antigen, DR alpha chain [Bos mutus]	Bos mutus	491	491	81%	2e-173	98.83%	257	ELR45153.1
☒	MHC class II antigen [Bos grunniens]	Bos grunniens	489	489	80%	4e-173	100.00%	253	AFL71083.1
☒	MHC class II antigen [Bos grunniens x Bos taurus]	Bos grunniens x Bos...	484	484	80%	7e-171	99.21%	253	AFC17944.1
☒	major histocompatibility complex, class II, DR alpha precursor [Bos taurus]	Bos taurus	482	482	80%	3e-170	98.81%	253	NP_001012695.1
☒	PREDICTED: HLA class II histocompatibility antigen, DR alpha chain [Bos indicus]	Bos indicus	481	481	80%	2e-169	98.42%	253	XP_019841331.1
☒	MHC class II antigen [Bos indicus]	Bos indicus	479	479	79%	8e-169	98.80%	251	CAX17687.1
☒	Major histocompatibility complex, class II, DR alpha [Bos taurus]	Bos taurus	479	479	80%	8e-169	98.02%	253	AAI02054.1
☒	MHC class II antigen [Bos indicus]	Bos indicus	474	474	80%	4e-167	97.23%	253	AOS51159.1
☒	mamu class II histocompatibility antigen, DR alpha chain-like precursor [Bubalus bubalis]	Bubalus bubalis	474	474	80%	6e-167	96.44%	253	NP_001277874.1
☒	MHC class II antigen [Ovis aries]	Ovis aries	464	464	80%	7e-163	95.65%	253	CAX17684.1
☒	MHC class II antigen [Ovis aries]	Ovis aries	462	462	80%	2e-162	95.26%	253	ALI94040.1
☒	MHC class II antigen [Ovis aries]	Ovis aries	462	462	80%	4e-162	95.26%	253	AFN25704.1
☒	mamu class II histocompatibility antigen, DR alpha chain [Oryx dammah]	Oryx dammah	461	461	80%	7e-162	94.07%	253	XP_040106323.1
☒	MHC class II antigen [Ovis canadensis]	Ovis canadensis	461	461	80%	9e-162	95.26%	253	AEX28226.1
☒	HLA class II histocompatibility antigen, DR alpha chain precursor [Capra hircus]	Capra hircus	460	460	80%	2e-161	94.86%	253	NP_001301117.1
☒	BoLA-DR-alpha [Bos taurus]	Bos taurus	459	459	75%	2e-161	99.58%	238	BAA07172.1
☒	MHC class II antigen [Ovis canadensis]	Ovis canadensis	460	460	80%	3e-161	94.86%	253	AEX28232.1
☒	mamu class II histocompatibility antigen, DR alpha chain-like precursor [Ovis aries]	Ovis aries	459	459	80%	4e-161	94.86%	253	NP_001267646.1

Background reading



Chapter 4 – Basic Local Alignment Search Tool (BLAST)

Chapter 5 – Advanced Database Searching